

# Hyunsoo Kim

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## EDUCATION

### The University of Texas at Austin

Austin, TX

*Ph.D. in Electrical and Computer Engineering*

*Sep 2026 – Present*

- Co-advised by Prof. Guandao Yang and Prof. Diana Marculescu
- Research: Leveraging image and video generation models for visual understanding and perception

### Korea University

Seoul, Republic of Korea

*M.S. in Artificial Intelligence*

*Sep 2024 – Aug 2026*

- Co-advised by Prof. Donghyun Kim and Prof. Suhyun Kim
- Research: Diffusion models, data-free synthesis, AI safety, and generative model alignment

### Inha University

Incheon, Republic of Korea

*B.S. in Statistics and Software Engineering*

*Mar 2018 – Aug 2024*

- Two-year break for military service (Jul 2019 – Feb 2021)

## RESEARCH INTERESTS

**Image and video generation** (diffusion models, flow matching, world models)

**Generate to Understand** (leveraging generative models for perception, reasoning, and visual understanding)

**AI Safety** (safe generation, concept erasure, trustworthy AI)

## PUBLICATIONS

- [C5] **Consistency-Preserving Concept Erasure via Unsafe-Safe Pairing (PAIR)** NeurIPS 2026
- Yongwoo Kim\*, Sungmin Cha\*, **Hyunsoo Kim**, Jaewon Lee, Donghyun Kim (\*: equal contribution)
  - Proposed PAIR, a consistency-preserving concept erasure framework that reframes erasure as semantic realignment using unsafe-safe pairs, with Fisher-weighted DoRA initialization for selective suppression while maintaining generation quality
- [C4] **Correlation-Weighted Multi-Reward Optimization for Compositional Generation** ECCV 2026
- Jungmyung Wi, **Hyunsoo Kim**, Donghyun Kim
  - Proposed correlation-based reward reweighting for multi-concept text-to-image generation; dynamically estimates which concepts conflict across generated samples and assigns them stronger optimization signals via GRPO fine-tuning on SD3.5 and FLUX.1-dev
- [C3] **Self-Improving Diffusion Classifiers with Minority Preference Optimization** ECCV 2026
- **Hyunsoo Kim**\*, Jungmyung Wi\*, Soobin Um, Donghyun Kim<sup>†</sup>, Suhyun Kim<sup>†</sup> (\*: equal contribution, †: equal advising)
  - Introduced MiPO, a self-improving framework that expands diffusion model coverage in low-density regions via GRPO with a novel DDIM-reconstruction-based minority preference reward
  - Improved zero-shot diffusion classification by up to 3.8% across multiple benchmarks without any additional real image data
- [C2] **Diffusion-assisted Data-free Image Synthesis with Domain-Class Alignment** ICML 2025
- Yujin Kim\*, **Hyunsoo Kim**\*, Hyunwoo J. Kim, Suhyun Kim (\*: equal contribution)
  - Proposed DDIS, a framework leveraging text-to-image diffusion models for data-free knowledge distillation with explicit alignment of domain and class distributions
- [C1] **Difference Inversion: Interpolate and Isolate the Difference with Token Consistency** CVPR 2025
- **Hyunsoo Kim**, Donghyun Kim<sup>†</sup>, Suhyun Kim<sup>†</sup> (†: equal advising)
  - Proposed a token-consistency-based DDIM inversion method that interpolates and isolates visual differences between image pairs, enabling image analogy generation for style transfer, object manipulation, and scene editing

## EXPERIENCE

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### Korea Institute of Science and Technology (KIST)

Seoul, Republic of Korea

*Graduate Researcher (Advisor: Suhyun Kim)*

*Sep 2024 – Aug 2026*

- Led research on image analogy generation, safe image translation, and self-improving diffusion classifiers
- Published 2 first-author papers at ECCV 2026 and CVPR 2025; 1 co-first-author paper at ICML 2025
- Developed training pipelines for diffusion model fine-tuning with RLHF/GRPO on multi-GPU clusters

### Korea Institute of Science and Technology (KIST)

Seoul, Republic of Korea

*Undergraduate Research Intern (Advisor: Suhyun Kim)*

*Mar 2024 – Jun 2024*

- Researched data-free image synthesis by optimizing text embeddings on text-to-image diffusion models
- Contributed to DDIS framework that was later accepted at ICML 2025

### Medical Image Innovation Lab, Seoul National University

Seoul, Republic of Korea

*Undergraduate Intern (Advisor: Wonjin Lee)*

*Dec 2023 – Feb 2024*

- Researched conditional generation and inpainting in medical imaging using diffusion models
- Implemented and evaluated diffusion-based inpainting pipelines for CT and MRI reconstruction tasks

## TECHNICAL SKILLS

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**Programming:** Python, C/C++, MATLAB, Bash, R

**ML Frameworks:** PyTorch, Hugging Face (Diffusers, Accelerate, TRL), DeepSpeed, JAX

**Tools:** Git, Docker, Weights & Biases, SLURM, Linux, L<sup>A</sup>T<sub>E</sub>X, Jupyter

**Research Areas:** Diffusion Models, Flow Matching, RLHF/GRPO, Image/Video Generation, Zero-shot Classification, AI Safety, Concept Erasure, Data-free Knowledge Distillation